Intelligence Beyond the Human

A Foundational Paper for SymK

SymK

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Scope: Human-facing explanation of the accepted Intelligence concept, its evidence, boundaries, open questions, and engineering consequences

> This paper is an explanatory and epistemic source. Publication does not make it a constitutional norm, canonical concept package, formal ontology, schema, assessment standard, authority allocation, or implementation specification. Exact Stage-Accepted decisions control where this paper and those decisions differ. The v0.1 publication remains part of the historical record.

Abstract

SymK began as a protocol for symbiotic cooperation between Human Intelligence and Artificial Intelligence, developed Domain Intelligence Amplifiers as a principal realization, and broadened into a responsible Knowledge Engineering framework. In that layered identity, Intelligence is foundational but not sovereign. It names a capacity; it does not by itself confer correctness, Knowledge, Agency, Authority, Responsibility, rights, moral standing, or permission to act.

The paper retains the Copernican correction of v0.1: Human Intelligence is an important manifestation and source of evidence, not the definitional center of all Intelligence. It also retains substrate independence without substrate indifference. Intelligence is not identical to biological tissue, hardware, software, a population, or an institution, yet every actual Intelligence is realized and borne by an organized system whose substrate, organization, dependencies, history, and failure modes matter.

The accepted SymK minimum is:

> **Intelligence is a substrate-realized, multidimensional capacity of an organized system to integrate relevant resources in producing context-sensitive interpretations or effective responses across materially varying situations, including cases not individually specified in advance.**

This definition separates the actual capacity from its manifestations and from fallible claims about it. Every actual Intelligence is borne by an organized system. A capability configuration is the Bearer’s actual multidimensional arrangement. An attributed profile represents that configuration for a declared purpose and can be incomplete, mistaken, outdated, or lossy. A performance is evidence under conditions, not the capacity itself. A score or class is a declared Projection, not a natural rank. An attribution record does not create the capacity it describes.

Knowledge and Intelligence remain independently defined and reciprocally related. Learning, adaptation, objectives, action, rationality, Agency, autonomy, consciousness, understanding, wisdom, Authority, Responsibility, origin, status, and moral or legal standing are not universal entailments of Intelligence. Collective, distributed, institutional, and hybrid cases are possible only under a justified Bearer boundary and system-level evidence; connection, aggregation, cooperation, or one improved result does not manufacture a new Intelligence.

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1. Why Intelligence is foundational to SymK

Intelligence appears in SymK's historical identity, its account of participants and capability, and its Domain Intelligence Amplifier orientation. Leaving it as a compliment, marketing category, Human-resemblance test, or benchmark score would create predictable errors:

1. Human resemblance could become an undeclared universal criterion.
2. A substrate, component, model, institution, or record could be mistaken for the capacity it supports or describes.
3. A successful output could be mistaken for stable capability.
4. A score could be mistaken for the whole multidimensional configuration.
5. accumulated Knowledge could be mistaken for the capacity that acquired, interpreted, or used it.
6. capability could be allowed to manufacture Authority, Responsibility, autonomy, consciousness, moral worth, or permission.
7. a collection could be called intelligent without evidence of a system-level Bearer.
8. Domain-specific findings could silently enter the universal Core.

SymK therefore needs a general concept plus disciplined attribution. The concept states a minimum. An attribution identifies a proposed Bearer, Scope, Context, relevant variation, Evidence, uncertainty, alternatives, temporal validity, and intended use. Neither definition nor attribution settles every question about mind, experience, life, ethics, law, or governance.

1.1 Layered project identity

The historical Human–AI protocol, the accepted Domain Intelligence Amplifier orientation, and the broader responsible Knowledge Engineering framework are layers of one project history. None should erase the others.

- The historical protocol explains the original cooperation problem and Human standing.
- The DIA remains a principal but non-exhaustive realization.
- The broader framework governs Knowledge Engineering beyond a fixed participant dyad.

The organizing center is the governed knowledge undertaking: a non-reifying analytical expansion over relevant Questions, participants, contributions, Knowledge, Reasoning, Answers, authority, Responsibility, Value, outcomes, consequences, feedback, correction, and Learning. It is not automatically a Foundational concept, System, Entity, Bearer, sovereign actor, package, or runtime object.

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2. The Copernican correction

2.1 Human Intelligence is an instance, not the center

> **Human Intelligence is a manifestation of Intelligence. Intelligence is not defined as resemblance to a Human being.**

Human language, embodiment, development, culture, sociality, affect, history, and mortality provide indispensable evidence. They do not thereby become necessary conditions for every Intelligence. A Human reference case can guide evaluation without becoming an essence.

Non-human biological cognition should not be described as deficient humanity. Comparative research reports plural, ecologically situated capacities and also warns that a psychometric general factor requires validation across species [Burkart, Schubiger, and van Schaik 2017]. Artificial systems should likewise be evaluated by identified capacity, realization, Scope, conditions, and Evidence rather than by theatrical Human imitation alone. Turing's operational turn remains important, but Human indistinguishability is not SymK's universal criterion [Turing 1950].

2.2 Decentering is not equivalence

Rejecting Human privilege at the level of definition does not establish equal capability, breadth, reliability, experience, consciousness, trust, Authority, Responsibility, moral standing, legal status, or rights. Those are separate claims requiring their own Grounds, Evidence, jurisdiction, and procedure.

The correction is methodological: origin and resemblance do not decide Intelligence in advance. It does not erase Human dignity, lived experience, affected-subject standing, external law, institutional authority, or the consequences borne by people.

2.3 Two senses of social Intelligence

An individual's capacity to interpret and navigate social situations is an Intelligence specialization. A coordinated social organization's possible system-level Intelligence is a Bearer claim. The first does not establish the second. Neither label is accepted merely because a person is socially successful or a group produces an aggregate result.

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3. The conceptual architecture

The actual phenomenon and the claims made about it must be separated.

3.1 Realization and manifestation

```
substrate and dependencies
       ↓ realize
organized system
       ↓ actually bears
multidimensional Intelligence configuration
       ↓ manifests under declared conditions
performance, interpretation, or response
```

3.2 Attribution and governance

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observations, tests, failures, interventions, and counterevidence
       ↓ support or challenge
Intelligence attribution claim about a proposed Bearer
       ↓ is evaluated through
assessment for a declared use
       ↓ may be represented by
profile, score/class Projection, status, authorization, and record
```

The sequences interact but are not identical. A system can bear Intelligence before recognition or recording. A confident record can be wrong. An approved use can be narrow. Withdrawal of authorization does not by itself remove the capacity. A later model, member, tool, or environmental change may alter either the Bearer, its configuration, or the validity of an older attribution.

3.3 Substrate

The substrate is the material, biological, computational, social, institutional, or hybrid realization base. Substrate affects latency, energy, embodiment, access, persistence, memory, modifiability, vulnerability, inspection, and failure. Substrate alone neither proves nor disproves Intelligence.

3.4 Organized system and Bearer

Organization—not mere collection—supports the capacity. Relevant organization may include architecture, feedback, memory, communication, division of labor, control, representations, interfaces, incentives, and temporal continuity.

Bearer is a relational role: what actually bears the capacity. Every actual Intelligence must be borne by an organized system, but the universal concept need not name a particular Bearer. An attribution identifies a proposed Bearer and fallibly claims that the bearing relation obtains.

The boundary must remain stable within the claim. Tools, retrieval, documents, sensors, people, models, institutions, and communication channels cannot move opportunistically inside or outside the Bearer merely to protect a favorable result.

3.5 Actual configuration and attributed profile

The Bearer's actual capability configuration is the multidimensional arrangement it has. A profile is a Representation of that configuration for a purpose. A profile can select dimensions, compress variation, omit dependencies, age, or be wrong. It never creates the capacity.

Candidate dimensions can include discrimination, perception, memory, integration, interpretation, inference, model formation, problem solving, planning, learning, transfer, coordination, creation, communication, error detection, metacognition, or objective revision. This list is an evidence-informed inventory, not a closed universal schema. The accepted minimum requires multidimensionality and integration; it does not accept every named dimension as Foundational or universally required.

3.6 Performance

Performance is an event or result under particular conditions. It depends on task, environment, Knowledge, training, tools, collaborators, prompts, time, computation, incentives, access, noise, and measurement. Success may result from durable capacity, narrow skill, memorization, leakage, luck, hidden assistance, environmental regularity, or a measurement defect. Failure may reflect absent capacity or blocked manifestation.

Repeated and materially varied evidence is therefore essential. Counterexamples, failures, component removal, resource changes, transfer tests, and time are often more informative than one demonstration.

3.7 Assessment, authority, authorization, status, and record

An assessment evaluates an attribution for a declared use. Competence to assess does not automatically create decision Authority. Authority requires a legitimate source, Scope, jurisdiction, and procedure. Authorization permits a bounded use; it is not the capacity. Responsibility remains separately allocated. A status or record represents an outcome of governance and can be challenged, superseded, restricted, or obsolete.

SymK preserves eleven typed Authority kinds under the accepted baseline. Affected-subject standing/challenge remains a distinct unresolved locus. It is not a settled twelfth Authority kind and does not automatically confer veto, legal standing, participation, consent, right, duty, remedy, or decision power.

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4. Substrate independence without substrate indifference

> **The universal concept of Intelligence does not contain one biological, electronic, social, or hybrid material as a necessary condition.**

This is a research commitment, not a proof that every capacity can be reproduced in every substrate. Higher-level comparison remains useful only when realization-specific mechanisms, dependencies, and failures are preserved.

Marr's levels of analysis show why a capacity cannot be explained by a component list alone [Marr 1982/2010]. Multiple-realizability arguments motivate comparison across mechanisms without requiring SymK to accept every associated philosophical conclusion [Putnam 1967; Fodor 1974]. Embodied, extended, and distributed traditions show why substrate and system boundary cannot be ignored [Varela, Thompson, and Rosch 1991; Clark and Chalmers 1998; Hutchins 1995].

Functional similarity in a declared Scope does not entail identical mechanism, generality, robustness, development, experience, consciousness, moral standing, Authority, or appropriate governance. The balancing formula is:

> **Substrate-independent in definition; substrate-sensitive in realization and Evidence.**

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5. What the research traditions contribute

The literature supplies partial projections and competing emphases, not one definition that SymK can copy unchanged.

> **Accessible table projection:** Each source row follows as a labelled record; all source headers and cell wording are preserved.

- **Tradition:** psychometric *g* and hierarchical ability models
  - **Primary evidence/object:** Human test covariance
  - **Durable contribution:** latent-variable discipline; broad and narrow structure
  - **Limit as universal definition:** population-, task-, culture-, and instrument-sensitive
  - **SymK use:** Human capability evidence and assessment design
- **Tradition:** plural, practical, emotional, and social constructs
  - **Primary evidence/object:** Domain-sensitive Human performance
  - **Durable contribution:** profiles can preserve important differences hidden by one scalar
  - **Limit as universal definition:** construct boundaries, overlap, and validation vary
  - **SymK use:** candidate dimensions or specializations, not admission by name
- **Tradition:** developmental and sociocultural theories
  - **Primary evidence/object:** change across development and mediation
  - **Durable contribution:** developmental history, instruction, tools, transfer
  - **Limit as universal definition:** primarily Human developmental models
  - **SymK use:** formation and temporal evidence
- **Tradition:** rational-agent and symbolic traditions
  - **Primary evidence/object:** action, objectives, search, representation
  - **Durable contribution:** explicit environment, objective, inference, and consequence models
  - **Limit as universal definition:** may collapse Intelligence toward Agency or optimization
  - **SymK use:** bounded decision and reasoning projections
- **Tradition:** universal-machine Intelligence
  - **Primary evidence/object:** policy performance across formal environments
  - **Durable contribution:** breadth, goals, environments, priors
  - **Limit as universal definition:** idealization and reward-centered assumptions
  - **SymK use:** theoretical generality projection
- **Tradition:** skill-acquisition efficiency
  - **Primary evidence/object:** performance relative to priors and experience
  - **Durable contribution:** separates skill from acquisition/generalization efficiency
  - **Limit as universal definition:** depends on declared Scope, priors, and benchmark construction
  - **SymK use:** disclosure and learning-efficiency evidence
- **Tradition:** embodied, enactive, extended, and distributed cognition
  - **Primary evidence/object:** coupled organism/system and environment

    - **Durable contribution:** realization and boundary matter
    - **Limit as universal definition:** boundary and constitutive claims remain disputed
    - **SymK use:** mechanism and Bearer inquiry
- **Tradition:** collective Intelligence
    - **Primary evidence/object:** coordinated group performance
    - **Durable contribution:** possible system-level capacity beyond strongest member
    - **Limit as universal definition:** aggregation can be mistaken for integration
    - **SymK use:** collective-Bearer evidence, not automatic attribution

Spearman, Cattell, Carroll, Thurstone, Wechsler, Gardner, Sternberg, and related traditions show that common variance, broad capacities, narrow abilities, and practically salient domains can coexist as empirical questions [Spearman 1904; Cattell 1943; Carroll 1993; Thurstone 1938; Wechsler 1944; Gardner 1983; Sternberg 1985]. Work testing multiple-intelligences claims likewise cautions against assuming independent natural kinds merely from named domains [Visser, Ashton, and Vernon 2006]. Emotional-Intelligence research illustrates the difference between ability constructs and broader mixed models [Salovey and Mayer 1990; Mayer, Salovey, and Caruso 2004].

Piaget and Vygotsky make development, mediation, and history visible [Piaget 1952; Vygotsky 1978]. Newell and Simon and the rational-agent tradition clarify constructive and decision-centered accounts without establishing them as the universal minimum [Newell and Simon 1976; Russell and Norvig 2020]. Legg and Hutter formalize goal achievement across environments [Legg and Hutter 2007]. Chollet emphasizes priors, experience, novelty, and skill-acquisition efficiency [Chollet 2019]. Woolley and collaborators provide evidence of group-level performance structure without proving that every group is a literal Intelligence [Woolley et al. 2010]. Recent formal proposals remain research inputs rather than constitutional authority [Ng 2025].

No literature source settles SymK semantics. Accepted decisions control the project meaning; the literature supplies Evidence, alternatives, cautions, and tests.

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6. The accepted definition

> **Intelligence is a substrate-realized, multidimensional capacity of an organized system to integrate relevant resources in producing context-sensitive interpretations or effective responses across materially varying situations, including cases not individually specified in advance.**

6.1 Substrate-realized

The capacity is realized through a physical, biological, computational, social, institutional, or hybrid organization. It is neither identical to the substrate nor free-floating.

6.2 Multidimensional capacity

Intelligence is more stable than one output and cannot be exhausted by one scalar. Dimensions may vary in strength, breadth, reliability, integration, development, and failure. The definition does not require one fixed universal dimension list.

6.3 Organized system

The Bearer is the level whose organization causally supports the claimed capacity. An organism, artificial system, team, institution, distributed arrangement, or hybrid can be proposed; none qualifies by label or origin.

6.4 Integrate relevant resources

Resources can include internal structure, memory, information, Knowledge, tools, participants, representations, energy, computation, and environmental access. Relevance is Context- and Scope-sensitive. Resource dependence does not disqualify a Bearer, but dependencies and boundary choices must be disclosed.

Knowledge is not inserted as a constitutive universal ingredient. An Intelligence may create, use, ignore, lack, or be misattributed Knowledge. Whether every actual Intelligence bears some practical Knowledge remains a live question.

6.5 Context-sensitive interpretations or effective responses

Intelligence need not be reduced to outward action, one objective, or one utility function. Interpretation and response can include explanation, inference, problem solving, planning, creation, coordination, or other capability manifestations. “Effective” is relative to a declared task or standard; it does not mean true, good, legitimate, authorized, responsible, or valuable.

6.6 Materially varying situations

Capacity must survive relevant variation rather than one repeated condition. What counts as material depends on the claim, Domain, environment, and failure horizon. Breadth, depth, transfer, learning efficiency, reliability, and resource use must remain distinguishable.

6.7 Cases not individually specified in advance

The clause separates flexible capacity from exhaustive lookup or direct enumeration. It does not require online learning, Human-like novelty, open-ended adaptation, self-set objectives, or autonomy. A fixed but flexible system can qualify within an honest Scope.

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7. Dimensions, specializations, configurations, profiles, and scores

An **Intelligence dimension** is one aspect along which relevant capability can be described or assessed. An **Intelligence specialization** constrains the universal concept to a declared Domain or family of situations and identifies relevant dimensions, variation, Evidence, limitations, and assessment conditions. An **actual configuration** is the Bearer’s real multidimensional arrangement. An **attributed profile** is a fallible Representation of that arrangement. A **score or class** is a Projection for a declared purpose.

These are not interchangeable:

- a named classification does not create a specialization;
- a specialization is not automatically a separate Foundational Intelligence;
- a profile does not create or exhaust an actual configuration;
- a score does not create rank outside its declared Projection;
- one strong dimension does not entail general superiority;
- overlap or covariance does not erase useful specialization;
- weakness in one dimension does not automatically negate a narrower supported attribution.

Emotional, social, logical-mathematical, biological, artificial, collective, distributed, and hybrid labels therefore require different treatment. Emotional, social, and logical-mathematical usually name candidate specializations or dimension bundles. Biological, artificial, social-system, distributed, and hybrid often describe realization or organization. None is a closed natural taxonomy by naming alone.

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8. Distinctions and prohibited entailments

8.1 Intelligence and Knowledge

Knowledge and Intelligence are independently defined and reciprocally related. Intelligence can contribute to acquiring, testing, interpreting, transforming, or applying Knowledge. Knowledge can scaffold, preserve, constrain, or sometimes amplify Intelligence. Neither is reducible to the other, and neither attribution automatically entails the other.

8.2 Intelligence, cognition, competence, and skill

Cognition is a wider process family. Skill is developed ability in a task class. Competence is capability relative to a declared practice or standard. Intelligence can support, integrate, transfer, or revise skills; accumulated skill can also reflect extensive priors and practice. Skill and competence are evidence-sensitive neighbors, not substitutes.

8.3 Intelligence and performance

Performance is manifestation under conditions. Capability attribution is an inference from patterns of performance, failures, variations, resources, and counterfactual evidence. Neither one success nor one failure is decisive by itself.

8.4 Intelligence and Learning or adaptation

Learning and adaptation can change a capability configuration and provide important Evidence. They are not universal constitutive requirements. Situational response, online Learning, development, evolution, external redesign, component replacement, and institutional change must not be collapsed.

8.5 Intelligence and Agency, autonomy, or rationality

Intelligence does not entail initiating or controlling action, setting objectives, operating independently, or responding correctly to reasons. Agentic, autonomous, or rational behavior requires separate attribution. A capable system may be advisory, constrained, biased, deceptive, misinformed, or directed toward harmful objectives.

8.6 Intelligence and understanding, consciousness, or wisdom

Behavioral or syntactic success has not settled understanding [Searle 1980]. Functional capacity does not settle subjective experience. Capability does not settle sound judgment about values or consequences. Understanding, consciousness, and wisdom require separate inquiry and Evidence.

8.7 Intelligence and status, Authority, Responsibility, or moral worth

Origin and Intelligence do not confer epistemic privilege, decision Authority, authorization, operational control, Responsibility, legal status, moral standing, dignity, rights, duties, liability, blame, remedy, or permission. These require distinct Grounds and competent procedures. Human dignity is not graded by an Intelligence score.

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9. Collective, distributed, institutional, and hybrid Intelligence

A collection does not become a new Bearer because its members interact, cooperate, share a tool, vote, concatenate outputs, or achieve one better result. A system-level claim should identify at least:

- the proposed system boundary and level;
- persistent organization or controlled continuity;
- integration, memory, feedback, correction, or differentiated interdependence;
- dependencies and control surfaces;
- performance across relevant variation;
- failures and component-removal or substitution evidence;
- explanatory advantage over member-level aggregation alone;
- and the temporal conditions under which the attribution remains valid.

The lowest adequate Bearer should be preferred: attribute the capacity to the least expansive organized system that explains the evidence without losing essential integration. Nested attributions are possible when Evidence supports different capacities at component and system levels.

Cooperation is an actual, scoped relational process in which two or more distinguishable Participants make materially relevant Contributions that are mutually oriented toward a bounded shared undertaking and sufficiently connected for the Contributions to form part of that undertaking rather than merely coincide or aggregate. It may combine complementary capacities, compensate for failures, or contribute to a new system-level capability. It may also produce interference, dependency, fragility, Power concentration, or loss. Cooperation therefore does not entail collective Intelligence, Amplification, improvement, consensus, ethical adequacy, Value, or Authority.

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10. Amplification and Domain Intelligence Amplifiers

Amplification is not more content, one improved output, one higher score, automation, substitution, or replacement. An amplification claim compares a target capability profile across a declared baseline and horizon, preserves gains, losses, unchanged dimensions, dependencies, integration effects, uncertainty, and attribution level, and survives relevant variation.

“Amplified Intelligence” is a profile-change claim, not a new substance or automatic status. The amplified Bearer may be a participant, a subsystem, or a newly integrated system; it must not be assumed.

A Domain Intelligence Amplifier is a principal SymK realization only when the accepted seven-condition test is met: (1) an identified organized System; (2) a declared Domain purpose and complex-question orientation; (3) plural distinguishable qualified Intelligence participants; (4) actual Cooperation among them; (5) governed Domain Knowledge as epistemic medium; (6) an explicit amplified-Bearer and capability target; and (7) evidenced system capability to support repeatable Amplification across relevant variation. The amplified Bearer need not be the DIA System. Authority, Responsibility, Value, outcome, consequence, challenge, lifecycle, and revision remain separately governed. The DIA is principal but not exhaustive of the broader framework.

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11. Three Views and an attribution review envelope

Human, Scientific, and Engineering Views are the universal minimum for expressing material Intelligence claims:

- the **Human View** makes the Bearer, purpose, meaning, limitations, consequences, uncertainty, and affected perspectives intelligible;
- the **Scientific View** states constructs, hypotheses, evidence, methods, alternatives, counterexamples, uncertainty, and limits;
- the **Engineering View** makes boundaries, interfaces, dependencies, observability, versioning, tests, failure modes, and control implications inspectable.

These Views are epistemic and representational functions, not file formats or stakeholder classes. A readable page, valid experiment, or precise schema does not by itself satisfy all three. Transformation among Views must declare source, target, Scope, omissions, assumptions, uncertainty, and material loss. No View or artifact owns meaning.

The v0.1 `IntelligenceObject` is therefore replaced by a non-container review envelope. For a declared claim and use, reviewers may inspect:

Accessible table projection: Each source row follows as a labelled record; all source headers and cell wording are preserved.

- **Area:** proposed Bearer and boundary
    - **Questions:** Which organized system is claimed to bear Intelligence? Why are these components inside?
- **Area:** realization
    - **Questions:** Which substrate, organization, interfaces, dependencies, and continuity conditions matter?

- **Area:** actual claim
  - **Questions:** Which capacity or specialization is claimed, in which Scope and Context?
- **Area:** profile and Projection
  - **Questions:** Which dimensions are represented? What is omitted? How is any score or class produced?
- **Area:** variation and time
  - **Questions:** Which situations vary materially? Which cases were not individually specified? How long is the claim valid?
- **Area:** priors and resources
  - **Questions:** Which inheritance, design, training, Knowledge, tools, assistance, computation, access, and permissions were available?
- **Area:** Evidence
  - **Questions:** Which observations, repeated tests, failures, interventions, counterfactuals, and alternative explanations matter?
- **Area:** assessment and uncertainty
  - **Questions:** Who assessed what for which use, by what method, with which uncertainty and disagreement?
- **Area:** authority and use
  - **Questions:** Which competent source may authorize reliance or operation in this jurisdiction?
- **Area:** Responsibility and consequences
  - **Questions:** Who remains responsible? Who benefits or is affected? Which outcomes and consequences return for review?
- **Area:** challenge and revision
  - **Questions:** Who may challenge claims or decisions? Which competent procedure changes an artifact, status, authorization, or authority source?
- **Area:** lineage
  - **Questions:** Which versions, component changes, supersessions, and unresolved dissent must remain visible?

The envelope is not a canonical class, package, record type, schema, workflow, certification, or universal admission gate. Its constituent roles remain distinct.

Evidence may challenge claims, premises, Applicability, consequences, and decisions operating under any Authority. It does not automatically amend a governed artifact, alter or revoke an Authority or its source, or reverse a decision; such change follows competent procedure. Urgent adverse Evidence triggers urgent consideration by competent operational or external Authority. This creates no emergency duty, containment or suspension power, decision right, remedy, or jurisdiction.

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12. Explanatory obligations

The v0.1 commitments I-01–I-12 are retained as explanatory obligations derived from accepted DR-009–018, not as candidate axioms or constitutional selections:

1. separate Intelligence from substrate, component, performance, profile, score, status, and record;
2. identify realization and relevant dependencies;
3. identify and justify the proposed Bearer and attribution level;
4. preserve the non-anthropocentric correction without asserting equivalence;
5. refuse capability or privilege by origin alone;
6. treat performance as conditional Evidence rather than identity;
7. preserve multidimensionality, Scope, variation, uncertainty, and declared Projection;

8. treat classifications and specializations as revisable, overlapping, evidence-sensitive constructs;
9. preserve Knowledge–Intelligence independence and typed reciprocity;
10. block automatic transfer to Learning, Agency, autonomy, rationality, understanding, consciousness, wisdom, Authority, Responsibility, or moral/legal status;
11. require more than aggregation for a collective or hybrid Bearer; and
12. preserve challenge, lineage, competent change procedure, and consequence return.

These obligations explain the accepted baseline. They do not Ratify constitutional wording, create a semantic package, or allocate actual authority.

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13. Counterexamples the account must survive

1. **Exhaustive lookup:** a table returns every enumerated answer correctly. Correctness alone does not show capacity across cases not individually specified.
2. **Fixed but flexible system:** a deterministic specialist handles new Domain cases without online Learning. It can qualify within Scope.
3. **Fluent one-off success:** an impressive output is Evidence, but not sufficient attribution.
4. **Narrow expert:** deep capability in one specialization need not imply broad generality.
5. **Coordinated aggregation:** voting or concatenation can improve results without creating a collective Bearer.
6. **Distributed institution:** durable organization, records, roles, feedback, and correction can support a system-level claim when the evidence warrants it.
7. **Tool-dependent Human:** resource dependence does not negate capacity; the claimed Bearer boundary must be explicit.
8. **Capable but unauthorized system:** a supported profile does not create decision Authority or permission.
9. **Stored profile:** a database record can be useful, false, or obsolete; it does not bear the capacity.
10. **More content without amplification:** access to documents and one better answer do not prove durable capability change.
11. **Uneven specialization:** strong logical performance and weak social interpretation require an honest multidimensional profile.
12. **Emotional classification without emotional Intelligence:** benchmark labels under narrow conditions do not establish the broader specialization.
13. **Aggregate gain with hidden loss:** a higher score can conceal declining robustness, error detection, or affected-subject sensitivity.
14. **Component replacement:** a model or member change can preserve, alter, or end the claimed Bearer; identity cannot be assumed.
15. **Evidence against an authorized use:** the attribution or decision may be challenged immediately, while artifact or authority-source amendment still requires competent procedure.
16. **Urgent adverse Evidence:** urgency triggers consideration by competent operational or external Authority; the Evidence itself creates no new duty, power, remedy, or jurisdiction.

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14. Development requirements

Implementations derived from this foundation should:

1. represent substrate, organized system, actual bearing, configuration, profile, performance, attribution, assessment, status, authority, authorization, Responsibility, and record distinctly;
2. keep a justified Bearer boundary stable across evaluation;
3. represent profiles before scalar scores and declare every Projection and material loss;
4. support overlapping dimensions and specializations without a flat natural-kind enumeration;

5. make Domain, Scope, Context, relevant variation, time, and version first-class;
6. disclose priors, resources, tools, assistance, dependencies, and permissions;
7. preserve successes, failures, alternatives, counterfactuals, uncertainty, and dissent;
8. support individual, collective, distributed, nested, institutional, and hybrid proposals without presuming them;
9. distinguish situational response, Learning, adaptation, development, evolution, redesign, and replacement;
10. prevent anthropomorphic, substrate-origin, fluency, benchmark, and authority leakage;
11. maintain all Three Views and declared cross-View loss;
12. separate Evidence challenge from competent artifact and authority-source change procedure;
13. preserve affected-subject standing/challenge as a distinct unresolved locus;
14. return outcomes and consequences to validation, Responsibility, policy, concept, and constitutional review where applicable;
15. treat schemas and records as Representations, not semantic authorities; and
16. evaluate amplification as durable multidimensional profile change, not generic increase.

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15. Accepted answers, live questions, and preserved dissent

Accessible table projection: Each source row follows as a labelled record; all source headers and cell wording are preserved.

- **v0.1 question:** minimum condition
  - **Accepted minimum:** exact multidimensional, integration, Context, material-variation, and non-individual-specification definition
  - **Live remainder:** thresholds and operational tests remain Domain-sensitive
- **v0.1 question:** whether Learning is necessary
  - **Accepted minimum:** no universal online-Learning requirement
  - **Live remainder:** Learning may be required by a specialization or assessment purpose
- **v0.1 question:** whether objectives are necessary
  - **Accepted minimum:** no universal fixed-objective requirement
  - **Live remainder:** objectives matter to some agentic and performance projections
- **v0.1 question:** how the Bearer is bounded
  - **Accepted minimum:** organized-system boundary, lowest adequate Bearer, integration and persistence Evidence
  - **Live remainder:** exact causal and identity criteria remain claim-sensitive
- **v0.1 question:** whether social systems can literally be intelligent
  - **Accepted minimum:** possible but not automatic
  - **Live remainder:** each claim requires system-level Evidence beyond aggregation
- **v0.1 question:** how much novelty is required
  - **Accepted minimum:** cases not individually specified and materially varying situations
  - **Live remainder:** operational distance measures remain Domain-sensitive
- **v0.1 question:** whether equivalent performance survives mechanism difference
  - **Accepted minimum:** equivalence is always scoped and does not transfer adjacent properties
  - **Live remainder:** robustness and mechanism-sensitive comparison remain empirical
- **v0.1 question:** relation to understanding
  - **Accepted minimum:** no automatic entailment
  - **Live remainder:** understanding may be a Knowledge mode, Intelligence dimension, both, or distinct
- **v0.1 question:** relation to consciousness

    - **Accepted minimum:** no automatic entailment
    - **Live remainder:** ethical inquiry is required if credible concern arises
- **v0.1 question:** persistent hybrid participant
    - **Accepted minimum:** possible under justified identity, integration, and continuity
    - **Live remainder:** component-change and succession criteria remain open
- **v0.1 question:** how claims age
    - **Accepted minimum:** temporal validity, versioning, challenge, restriction, and supersession are required
    - **Live remainder:** detailed aging rules belong to later standards
- **v0.1 question:** irreducible primitives
    - **Accepted minimum:** accepted Foundational concepts and supporting distinctions govern
    - **Live remainder:** final package anatomy, identifiers, and formal semantics remain downstream

Preserved dissent includes whether every actual Intelligence entails some practical Knowledge; which dimensions belong to the universal semantic inventory; when a named specialization is an Intelligence specialization rather than a skill or capability bundle; how profile comparison avoids Domain-specific value smuggling; when a collective or hybrid is literally intelligent; and which parts of attribution belong to universal semantics versus engineering standards.

These are not new deferred questions created by this paper. They preserve the accepted dissent and downstream ownership of DR-011–018.

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16. The branching value relation

The v0.1 motto “Knowledge is the shared medium. Cooperation is the objective” is historically important but too linear as a universal statement. The accepted relation is branching, defeasible, and recursive:

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Question → Reasoning → Answer
       ↘ evidence, challenge, and alternative branches ↗
Answer → possible use or decision → outcomes → consequences
Knowledge, Cooperation, capability, time, total cost, and Authority condition the branches
Consequences may return to validation, Responsibility, policy, concepts, or constitutional
review
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Knowledge can mediate the undertaking; Cooperation can be one process; amplified Intelligence can be a target capability; and better answers, faster and at lower total cost can be intended comparisons. None is guaranteed. No output, cooperation event, answer, outcome, or metric automatically creates truth, Value, improvement, authorization, or responsible operation.

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17. Conclusion

The Intelligence foundation is plural without being vague, substrate-independent without being substrate-indifferent, and non-anthropocentric without asserting equivalence.

Its conceptual center is exact:

> **Intelligence is a substrate-realized, multidimensional capacity of an organized system to integrate relevant resources in producing context-sensitive interpretations or effective responses across materially varying situations, including cases not individually specified in advance.**

Every actual Intelligence requires an organized system that bears it. Attribution of that Bearer is subsequent and fallible. An actual configuration is not its profile; a profile is not its score; a performance is not the capacity; a record is not the phenomenon; capability is not Authority; and origin is not status.

Human Intelligence remains a central participant and source of evidence without serving as the measure of every possible Intelligence. Artificial, biological, collective, distributed, institutional, and hybrid cases remain possible without receiving attribution by label, connection, fluency, cooperation, aggregation, publication, or score.

Knowledge and Intelligence remain distinct and reciprocal. Cooperation can contribute to new capability but does not guarantee it. Amplification is a comparative, scoped, temporal, and evidenced relation concerning a material contribution to increased effective capability at an identified organized Bearer relative to a materially comparable baseline across declared relevant variation. DIA remains a principal but non-exhaustive realization of the wider responsible Knowledge Engineering framework.

This v0.2 Markdown has completed Stream F cross-paper integration and is ready for later publication-projection work and complete Human reading. It is not yet a published edition. No PDF, accessibility acceptance, Human-comprehension result, constitutional Ratification, semantic package, or implementation Authority follows.

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Document governance note

This edition preserves the v0.1 publication and the early G/H review lineage while reconstructing the explanatory paper under exact DR-011–018. Stream F cross-paper integration is complete analytically through `SYMK-2X-EV-166`. Changes to literature characterization, conceptual wording, accepted-decision explanation, attribution guidance, diagrams, or downstream ownership must remain explicit and traceable.

No implementation should silently promote the dimensions, specializations, review envelope, obligations, counterexamples, or development inventory into a canonical schema. No Human review, PDF correspondence, visual/accessibility approval, or publication decision is claimed by this Markdown review draft.